

AI assistance and provenance

Repository source: AI_ASSISTANCE.md

Repository status: public-release construction checkpoint **Current named author:** Bruce J. Swihart **Primary AI system used:** OpenAI ChatGPT (GPT-5.6 Sol Pro) **Primary access period:** August-September 2026

Scope of AI assistance

OpenAI ChatGPT was used extensively during this project to:

- read and compare the supplied literature on bodies of constant width;
- discuss candidate proof strategies and failure modes;
- derive, reorganize, and check mathematical calculations;
- write and execute exact-rational Python verifiers;
- prepare an independently organized binary64 Python audit;
- prepare a matching base-R audit;
- design adversarial mutation tests;
- flatten the private proof dependency closure into a standalone public-format certificate;
- draft and revise the paper, Markdown documentation, build scripts, and continuous-integration workflow;
- inspect rendered manuscript and documentation PDFs; and
- organize claim boundaries, release gates, and reproducibility records.

The AI system is not listed as an author and is not represented as an independent reviewer. Its mathematical, computational, bibliographic, and expository output can be wrong. Every AI-proposed artifact intended for the repository is subject to deterministic checks and human review.

Human contribution and responsibility

Bruce J. Swihart selected the research problem and project goals, supplied and reviewed the source materials, chose which proposed directions to pursue, ran the independent base-R checks locally, maintained the private and fresh-history repositories, reviewed the manuscript and supporting documents, and decides whether and when a version is made public.

The named author is responsible for:

- the mathematical claims and their wording;
- the accuracy of the provenance and verification record;
- the final author list, affiliations, acknowledgments, and license;
- preserving corrections and superseded versions;
- not presenting the work as peer reviewed or independently verified before such review occurs; and
- ensuring that no private research history or unrelated private work is exposed in the public repository.

Relationship to the HYRA predecessor

The direct contact-geometric predecessor used by this project is the public HYRA artifact titled *A Certified Geometric Lower Bound for the Three-Dimensional Blaschke-Lebesgue Problem*. It is treated as an external source and cited as such. This repository does not claim authorship of that artifact. The current package develops and independently restructures its contact-geometric route, adds a new coordinate-width handoff, and provides a separate fresh-history verifier and release record.

Verification boundary

The repository contains several layers:

1. exact integer and rational verification of the finite certificate;
2. immutable machine-readable proof objects with SHA-256 fingerprints;
3. independent binary64 Python and base-R audits;
4. reconstruction, regression, and adversarial mutation tests;
5. automatic reconciliation of paper-facing constants against the certificate;
6. reproducible paper and documentation builds; and
7. human-readable derivations and claim ledgers.

These layers improve traceability and make discrepancies easier to detect. They do not turn an AI-assisted draft into an independently reviewed theorem. The current designation is:

AI-assisted, non-peer-reviewed computer-assisted research draft seeking independent mathematical verification.

AI output that is not proof authority

No natural-language answer from an AI system is a proof object. In particular, the following are not theorem authority:

- chat transcripts;
- hidden model reasoning;
- prose summaries of calculations;
- model confidence statements;
- web-search summaries;
- floating-point candidate searches; and
- AI-generated bibliography entries that have not been checked against primary sources.

The finite proof authority is described in [certificate/TRUST_BOUNDARY.md](#).

Model-identification note

The model designation records the hosted system used during this phase of the project. Hosted systems can change over time, and their behavior is not reproducible merely from a model name. The designation and access period are included for provenance, not as a claim that future model outputs will match those used here.

Disclosure in the paper and release

Before release, the paper and repository metadata should include a concise AI assistance disclosure consistent with this document. Any later material change in the role of AI systems should be recorded in a new version rather than silently rewriting the history of the released package.